

Math is Finding Patterns

Lesson 1-5

Name:

Date:

Class:

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

Language Objective: I can describe a pattern using the words pattern rule, generalization, and 'each number is ____ more than the one before it.'



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

pattern

pattern rule

generalization

table of values

reasonableness



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— Something that repeats or changes in a predictable way.

2

— The instruction that tells how to get from one number in a pattern to the next.

3

— A statement that is true for a whole pattern, made by noticing repeated calculations.

4

— A table that lists each step of a pattern so you can check that a solution is reasonable.

5

— Whether an answer makes sense — checked as you work, with adjustments made when needed.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

JUSTIFY

How can pattern rules tell you how many points each player has after the streak — and how can a table of values prove your answer is reasonable?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **pattern** — Something that repeats or changes in a predictable way.
patrón — Algo que se repite o cambia de una manera predecible.

- ☐ **pattern rule** — The instruction that tells how to get from one number in a pattern to the next.
regla del patrón — La instrucción que dice cómo pasar de un número del patrón al siguiente.

- ☐ **generalization** — A statement that is true for a whole pattern, made by noticing repeated calculations.
generalización — Una afirmación verdadera para todo un patrón, hecha al notar cálculos repetidos.

- ☐ **table of values** — A table that lists each step of a pattern so you can check that a solution is reasonable.
tabla de valores — Una tabla que muestra cada paso de un patrón para comprobar que una solución es razonable.

- ☐ **reasonableness** — Whether an answer makes sense — checked as you work, with adjustments made when needed.
sensatez del resultado — Si una respuesta tiene sentido — se comprueba mientras trabajas y se ajusta cuando es necesario.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Player A's pattern rule leads to a final score of ____ points. Player B's pattern rule leads to a final score of ____ points. Creating a ____ of values that checks each step confirms both totals are correct.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The lesson distinguishes patterns that keep going (like coin value scaling) from one-time facts (like the jar's specific total).

Evidence: A strong answer explains a pattern applies to any number of coins (it generalizes), while a fact like \$53.50 is fixed to this one jar. A weak answer just labels each card without explaining the difference.

Reasoning: This evidence connects the definition of pattern to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Player A's pattern rule leads to a final score of ____ points. Player B's pattern rule leads to a final score of ____ points. Creating a ____ of values that checks each step confirms both totals are correct.**
- **My answer is ____.**

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
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4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

